NPT STOCK INTELLIGENCE UNIT

Drug Price Intelligence

Proof of Concept Roadmap

4-Week Sprint Plan | Free Data Only | Live Predictions

4 Weeks to PoC	**50+ Molecules**	**£0 Data Cost**

March 2026 | Version 1.0 | CONFIDENTIAL

What We're Building in 4 Weeks

A working prototype that proves the core value: predicting which drugs are about to get expensive — using only free data.

50+

Molecules Tracked Live

High-risk generics from NHSBSA concession history

10

Free Data Sources

NHSBSA, OpenPrescribing, CPE, yfinance, BoE and more

>50%

Backtest Accuracy Target

Predict 2023–25 concession events using historical signals

PoC Definition of Done: A Streamlit dashboard, live on a local server, showing real-time risk scores and buying recommendations for 50+ molecules — with a backtesting report proving >50% historical prediction accuracy. Zero paid data required.

Free Data Sources — All £0 Cost

Every signal in the PoC comes from publicly available or free-tier APIs. Paid sources (WaveData) are deferred to Phase 2.

NHSBSA Open Data
T1+T2

Drug Tariff prices (historical + current), price concession records, dispensing data

CPE Website (scrape)
T2

Active price concession requests from pharmacies — a real-time distress signal

Bank of England API
T1

Official sterling exchange rates and base rate history — authoritative and free

MHRA / GOV.UK
T2

Public shortage notifications, GMP warnings, marketing authorisation status

pytrends (Google)
T2

Google Trends for condition/drug searches — early social demand signal, free Python library

OpenPrescribing API
T2

Monthly Rx volumes per BNF code — detect demand surges before they hit supply

yfinance (Yahoo)
T1

GBP/INR, GBP/CNY, energy futures, Brent crude — free Python library, daily refresh

NICE RSS Feeds
T1

Technology appraisal pipeline — upcoming drug approvals that will spike demand

ClinicalTrials.gov API
T1

Phase III trials by molecule — predicts new indications and demand surges ahead

EMA EPAR Database
T1

Biosimilar listing and reference biologic data — free EU regulatory database

Tier 1 Signal (3–12 months) | Tier 2 Signal (1–3 months) | Regulatory Intelligence

4-Week PoC Roadmap

Two sprints of two weeks each. Clear deliverable gate at end of every week. Full demo at end of Week 4.

WEEK 1 Data Foundation	WEEK 2 Signal Engine	WEEK 3 Prediction Model	WEEK 4 Dashboard & Demo
Set up n8n (self-hosted, free)	Unprofitable Floor Calculator	Historical model training (sklearn)	Streamlit dashboard (all screens)
NHSBSA Drug Tariff PDF parser	Generic manufacturer count tracker	Backtesting: 2023–25 concessions	Weekly intelligence email briefing
5-yr historical concession data pull	CPE website scraper (live concessions)	6-action buying recommendation logic	Full system integration test
OpenPrescribing API connector	NICE RSS feed watcher	Push/pull force weighting layer	Validated: 3+ live predictions
yfinance GBP/INR daily feed	MHRA shortage alert scraper	Patent cliff detector (manual + free APIs)	Demo to procurement team
SQLite database schema	Signal scoring engine (rules-based)	Alert threshold configuration	Gaps doc: what paid data unlocks
Define top-50 molecule watchlist	Tier 1 + Tier 2 signal dashboard	Model accuracy report (internal)	Phase 2 scope sign-off
✓ Pipelines live, data flowing	✓ All signals scoring, alerts set	✓ Model trained, backtest complete	✓ Live demo delivered ✓

Week 1 — Data Foundation

WEEK 1 / 4

rces flowing into a structured database. No intelligence yet — just clean, reliable data pipelines.

TASKS & DELIVERABLES

n8n self-hosted install — Data Eng

Deploy n8n Community on a local/team server. Free. All workflows run here.

NHSBSA Drug Tariff parser — Data Eng

Download current + 5 years of historical Drug Tariff PDFs. Parse to CSV. Store in SQLite.

NHSBSA Open Data concessions — Data Eng

Pull all historical price concession records from NHSBSA Open Data Portal API (free).

OpenPrescribing API connector — Data Eng

n8n workflow pulling monthly Rx volumes for top-50 BNF codes. Free REST API.

yfinance FX + Commodity feeds — Data Sci

Daily GBP/INR, GBP/CNY, Brent crude via yfinance Python library. Schedule via n8n Code node.

Database schema design — Data Eng

Define SQLite tables: molecules, tariff_prices, concessions, signals, recommendations.

Top-50 molecule watchlist — PM

Filter molecules most frequently appearing in 2021–25 NHSBSA concession records. Prioritise high-volume generics.

KEY DECISIONS & SPECS

Database:	SQLite (free, zero config)
Orchestration:	n8n Community (self-hosted, free)
Concession history:	NHSBSA Open Data API: nhsbsa.nhs.uk/opendata
Drug Tariff:	NHSBSA monthly PDF + Open Data CSV
Rx volumes:	openprescribing.net/api/1.0/spending/
FX rates:	yfinance: GBPINR=X, GBPCNY=X (free)
Molecule count:	Top 50 by concession frequency (2021–25)

Week 1 Gate: All data pipelines running. Molecule DB populated. Team can query live data.

Week 2 — Signal Engine

WEEK 2 / 4

ntelligence signals. By end of week, every molecule has a live risk score based on automated signal inputs.

TASKS & DELIVERABLES

Unprofitable Floor Calculator — Data Sci
Per-molecule: find historical price at which shortages began. Alert when DT price approaches.

Generic manufacturer count — Data Eng
Parse NHSBSA Part VIIIA monthly to count active suppliers per molecule. Flag when ≤2.

CPE concession scraper — Data Eng
Scrape CPE website for current price concession list. Store and compare month-over-month.

NICE RSS feed watcher — Data Eng
n8n HTTP Request node polling NICE appraisal RSS. Classify approvals vs rejections.

MHRA shortage alert scraper — Data Eng
Scrape MHRA and gov.uk for active shortage notifications and Serious Shortage Protocols.

Rules-based signal scoring — Data Sci
Weighted scoring engine combining all active signals into 0–100 risk score per molecule.

Tier 1 + 2 signal monitor view — Full Stack
Simple table/grid showing each molecule's signal statuses (green/amber/red). Internal use only.

SIGNAL WEIGHTS (RULES-BASED v1)

DaSH / Shortage Notifications	MHRA (free)	25%
Wholesaler OOS Frequency	Manual / CPE proxy	20%
CPE Concession Volume Trend	CPE scrape (free)	15%
Generic Manufacturer Count	NHSBSA Part VIIIA (free)	15%
DT Price vs Unprofitable Floor	NHSBSA + historical (free)	15%
GBP/INR + GBP/CNY Trend	yfinance (free)	5%
Seasonal Calendar Position	Built-in date logic	5%

Week 2 Gate: Every molecule has a live risk score. Alerts firing on threshold breaches.

Week 3 — Prediction Model & Backtesting

WEEK 3 / 4

odel on 5 years of historical data. Validate it against real concession events. Prove the prediction thesis.

TASKS & DELIVERABLES

Historical model training — Data Sci
scikit-learn GradientBoostingClassifier. Target: was price concession granted within 60 days? Train on 2020–24 data.

Backtesting: 2023–2025 — Data Sci
Replay historical signals for each molecule-month. Compare model predictions to actual NHSBSA concession records.

6-action buying recommendation — Data Sci
Rules engine mapping risk score + signal pattern → Buy Forward / Buffer / Diversify / Hold / Switch / Monitor.

Push/pull force weighting — Data Sci
Apply category-level multipliers from research (e.g. established generics 1.2x, branded 0.8x) to refine scores.

Patent cliff detector — PM + Data Sci
Use DrugPatentWatch free tier + manual research to flag molecules within 18 months of patent expiry.

Alert threshold tuning — Data Sci
Calibrate thresholds to minimise false positives. Target: ≤35% false positive rate. Adjust per molecule category.

Model accuracy report — Data Sci
Internal PDF report: precision, recall, F1 by category. Feed into stakeholder confidence-building conversation.

BACKTEST METHODOLOGY

1. **Pull ground truth**
   Download all NHSBSA price concession records 2020–2025. Each row = molecule + month + whether concession was granted.
2. **Replay signals historically**
   For each molecule-month, reconstruct what signal scores would have been at that time using archived data.
3. **Predict vs actual**
   Run model on historical signals. Did the model flag high risk ≥4 weeks before concession was granted?
4. **Measure accuracy**
   Precision, Recall, F1 per category. Report alongside false positive rate per signal weight combination.
5. **Find threshold**
   Identify the risk score cutoff that maximises useful alerts vs noise. This becomes the alert threshold for the PoC.

Week 3 Gate: Model trained. Backtest report showing ≥50% accuracy. Buying logic working.

Week 4 — Dashboard, Demo & Sign-Off

WEEK 4 / 4

rocurement will actually use. Validate live predictions. Demo to stakeholders and agree Phase 2 scope.

TASKS & DELIVERABLES

Streamlit dashboard — all screens — Full Stack
Home risk heatmap, Molecule Explorer, Signal Monitor, Buying Recommendations queue. Free Python library.

Weekly email briefing — Full Stack
Auto-generated summary email: top 5 at-risk molecules, active alerts, recommendations. Sent Mon 8am via n8n.

Full integration test — All
End-to-end test: data pipeline → signal scoring → model → recommendation → dashboard → alert. Fix all failures.

Validated live predictions — PM + Data Sci
Identify ≥3 molecules currently showing elevated risk. Document predictions. Track outcomes over next 8 weeks.

Procurement team demo — PM
45-minute demo to Head of Procurement + buyers. Walk through 3 real buying recommendations from the tool.

Free vs paid data gap analysis — PM + Data Sci
Document: which signals need WaveData? What do we gain? What is the accuracy improvement estimate?

Phase 2 scope sign-off — PM
Agree WaveData budget, wholesaler API access, and ML model upgrade plan. Produce Phase 2 roadmap draft.

STREAMLIT DASHBOARD SCREENS

Home — Risk Heatmap
Top 10 at-risk molecules, active alerts, signal status overview

Molecule Explorer
Search + sort all 50 molecules by risk score, category, recommendation

Molecule Detail Page
Price chart, floor distance, signal breakdown, buying action

Signal Monitor
All 57 signals with live status (green/amber/red) and last refresh time

Buying Recommendations
Action queue sorted by urgency — Buy Forward, Buffer, Monitor etc.

Alert Log
History of all fired alerts with molecule, signal trigger, and outcome tracking

Backtest Report
Model accuracy scorecard — builds stakeholder trust from Day 1

Week 4 Gate: Live demo delivered. Phase 2 scope agreed. Procurement team using tool daily.

PoC Tech Stack — 100% Free Tools

Every component is open-source or free-tier. No vendor lock-in. Easy to replace individual components in Phase 2.

DASHBOARD

Streamlit
Python web UI, free, runs locally or on free Streamlit Cloud

Plotly / Altair
Interactive charts in Python, both open-source

SMTP (n8n)
Weekly briefing email via n8n Email Send node

INTELLIGENCE LAYER

scikit-learn
GradientBoostingClassifier for concession prediction, 100% free

pandas + numpy
Signal scoring, floor calculation, data manipulation

Custom rules engine
Python: 6-action buying recommendation logic

ORCHESTRATION

n8n Community
Self-hosted workflow automation. Replaces £100s/mo of Zapier. Free forever.

Python Code Nodes
yfinance, pytrends, pandas all run inside n8n Code nodes

Schedule / Webhook nodes
Daily FX updates, monthly Drug Tariff parse, live CPE alerts

DATA STORAGE

SQLite
Zero-config, no server needed. Perfect for PoC scale (50 molecules).

CSV flat files
NHSBSA Drug Tariff data, historical concessions — stored as plain CSVs

PostgreSQL (Phase 2)
Migrate when scaling to 500+ molecules and multi-user access

Phase 2 upgrades: SQLite → PostgreSQL | Rules model → LightGBM ensemble | WaveData API (paid) | Cloud hosting | Wholesaler EDI feeds

Team & Roles — Who Does What

Minimum viable team for a 4-week PoC. Roles can overlap if headcount is limited — priorities are clearly defined.

Role	Week 1	Week 2	Week 3	Week 4	Alloc.
Product Manager / Lead *Ani*	Define molecule watchlist, data source audit, stakeholder alignment	Signal priority decisions, requirements sign-off	Backtest review, buying logic validation, procurement team prep	Demo delivery, Phase 2 scope, gap analysis, roadmap update	20%
Data Engineer *TBC*	n8n setup, NHSBSA parser, all data pipeline builds, SQLite schema	CPE scraper, MHRA scraper, NICE RSS, manufacturer count tracker	Historical data preparation, feature engineering, model pipeline	Integration testing, alert system hardening, email briefing	100%
Data Scientist / ML *TBC*	yfinance setup, floor calculator design, signal weight research	Rules-based scoring engine, signal normalisation, risk scoring	Model training, backtesting, threshold calibration, accuracy report	Live prediction validation, buying logic fine-tuning, documentation	100%
Full Stack / Frontend *TBC (or Data Eng)*	Local server setup, SQLite integration, basic query interface	Internal signal table view, alert notification skeleton	Dashboard wireframes, Streamlit prototype	All 7 dashboard screens live, email template, QA and fixes	60%

PoC Success Criteria — How We Know It's Working

The PoC is successful when all five criteria are met. Anything less is a learning — not a failure.

1

Data Pipelines 100% Automated

All 10 free data sources flowing into SQLite automatically via n8n. No manual downloads. Data freshness indicator shows all sources updated within expected refresh window.

✓ 0 manual data pulls

2

50+ Molecules Scored Live

Every molecule in the watchlist has a real-time risk score (0–100) calculated from live signal inputs. Scores update at minimum daily for Tier 2 signals.

✓ ≥50 molecules scored

3

Backtest Accuracy ≥50%

Running the model against 2023–2025 NHSBSA concession history: the model correctly predicts ≥50% of price concession events at least 4 weeks before they were granted.

✓ ≥50% precision/recall

4

3 Live Validated Predictions

At end of demo, the tool identifies ≥3 molecules currently at elevated risk. These are documented and tracked over the following 8 weeks to verify accuracy in the real world.

✓ 3 live calls documented

5

Procurement Team Using It

Following the Week 4 demo, at least 2 members of the procurement team access the dashboard independently and act on ≥1 buying recommendation within 2 weeks of launch.

✓ ≥1 action taken

PoC Risks & Mitigations

4 weeks is tight. Here are the most likely blockers and exactly how we handle each one.

NHSBSA historical data quality

L: Medium | I: High

PDF format changes or gaps in historical concession records may make backtesting unreliable.

↳ **Fix:** Cross-reference with CPE concession archive. Use NHSBSA Open Data CSV where available. Accept 3-year dataset if 5-year is incomplete.

No wholesaler OOS data (free)

L: High | I: Medium

AAH/Alliance OOS data requires wholesaler EDI access — not available for free in PoC.

↳ **Fix:** Use CPE price concession request volume as a strong proxy (pharmacies report OOS when requesting concessions). Flag as gap for Phase 2.

Model cold start: low accuracy

L: Medium | I: High

Backtest accuracy may come in below 50% on first run — this could undermine stakeholder confidence.

↳ **Fix:** Pre-select molecule watchlist to include high-frequency concession drugs where historical signals are richest. Tune on these first. Exclude hard cases from PoC scope.

n8n pipeline failures

L: Medium | I: Medium

Scrapers break if CPE or MHRA change their HTML structure. Scheduled jobs may fail silently.

↳ **Fix:** Build freshness indicators into dashboard — every data source shows last-updated timestamp. Alert when data age exceeds 2x expected refresh window.

Procurement team adoption

L: Medium | I: High

Team may not trust or use the tool if early predictions are wrong or interface is confusing.

↳ **Fix:** Involve 1 buyer as PoC tester from Week 2. Show backtest scorecard at demo. Focus Week 4 on the 3 live predictions — real wins build trust faster than accuracy metrics.

Scope creep

L: High | I: Medium

Stakeholders may request adding Scotland/Wales signals, OTC drugs, or branded medicine coverage in Week 1.

↳ **Fix:** Strict 50-molecule, England-only, NHS generics scope for PoC. Everything else goes into a backlog. PM holds the line.

From PoC to Production

What the PoC proves — and what paid data unlocks in Phase 2

WHAT THE POC PROVES

✓ Free data is enough to prove the thesis

NHSBSA + CPE + OpenPrescribing covers the most predictive signals (unprofitable floor, manufacturer count, concession volume).

✓ n8n can orchestrate all signals at scale

The automation architecture in the PoC carries directly into production — no rebuild needed.

✓ Rules-based model works for Phase 1

60–70% accuracy is achievable with rules alone. ML gives 80%+ — but that requires more training data and WaveData history.

✓ Streamlit is fine for internal tool MVP

Fast to build, easy to update, zero hosting cost. Upgrade to React/Next.js in Phase 3 if needed.

WHAT PAID DATA UNLOCKS (PHASE 2)

WaveData API

24 years of market prices → accurate unprofitable floor per molecule → +15–20% accuracy lift

Wholesaler EDI (AAH/Alliance)

Real-time OOS notifications → Tier 2 signal most predictive of 1–3 month events

DaSH Portal (DHSC)

6-month advance shortage warnings direct from manufacturers → Tier 1 critical signal

LightGBM ML model

Replace rules-based scoring. +10–15% accuracy over GradientBoosting with larger feature set